

```

1 * NOTE: idt() = integration function
2 .SUBCKT Block1_fixe_Drehzahl U1 V1 W1 F1 M n GND1
3 + PARAMS:
4 + UN_verkettet=20000
5 + IN=14.434k
6 + Pel=400e6 ;UN_verkettet*IN*sqrt(3)*cosphi of 0.8
7 + fn=50
8 + Kc=0.5
9 + UE0=600V ; Power of exciter = 0.25-5% * SN machine (smaller values for higher SN)
10 + IE0=2000A
11 + p=2
12 + Streuung=0.05 ;
13 + rcu=0.001
14 + rbg=0.001
15 + Schenkelpolfaktor=1 ;(1=cylindrical pole 0.5=Salient pole)
16 + Daempferfaktor=1
17 + err0=0
18 + RE_factor=0.6666
19 + n0=1500
20 + n1={n0} ;starts with 1500U/min
21 + alpha_0=0
22 + a=0.9 ; Parameter for nonlinear noload line (should be about 0.6-0.9)
23 + m=7
24 .PARAM
25 + a1={a}; Saturation parameter d-axis
26 + m1={m}; Saturation parameter d-axis
27 + a2={a1}; Saturation parameter q-axis
28 + m2={m1}; Saturation parameter q-axis
29 + UN={UN_verkettet/sqrt(3)}
30 + ZN={UN/IN}
31 + Rs={ZN*rcu}
32 + omega_el={(2*pi)*fn}
33 + Ls={Streuung*ZN/Kc/omega_el}
34 + PsiN={sqrt(2)*UN/omega_el}
35 + RE={UE0/IE0}
36 + RD={Rs/Daempferfaktor}
37 + RDx={RD} ; can be overwritten by setting this parameters on modelline
38 + RDy={RD} ; can be overwritten by setting this parameters on modelline
39 + LDx={Ls} ; can be overwritten by setting this parameters on modelline
40 + LDy={Ls} ; can be overwritten by setting this parameters on modelline
41 + LE={Ls} ; can be overwritten by setting this parameters on modelline
42 + R0={Rs} ; can be overwritten by setting this parameters on modelline
43 + L0={Ls} ; can be overwritten by setting this parameters on modelline (X0 is 0.03-0.1 p.u)
44 + Lsd={Ls}
45 + Lsq={Ls}
46 + XD={UN/IN/Kc}
47 + XQ={XD*Schenkelpolfaktor}
48 + g={UE0/(UN*sqrt(2)*RE/(XD-omega_el*Lsd))} ; updated formula for g parameter
49 + g1={UE0/(UN*sqrt(2)*RE/(XQ-omega_el*Lsq))} ; updated formula for salient pole machine
50 + ImNx={IE0/g} ; d-axis magnetizing current
51 + ImNy={IE0/g1} ;q-axis magnetizing current
52 + MN={Pel*(1-rbg-rcu)*p/omega_el}
53 *
54 *      currents of windings
55 G1  U1  U2  VALUE={v(isxS)+v(i_Null)}
56 G2  V1  V2  VALUE={{-v(isxS)+v(isyS)*sqrt(3))/2+v(i_Null)}
57 G3  W1  W2  VALUE={{-v(isxS)-v(isyS)*sqrt(3))/2+v(i_Null)}

```